

## MIHAIL KOGALNICEANU, Romania

WMO: 154810

Lat: 44.362N

Lon: 28.488E

Elev: 353

StdP: 14.51

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 12.1       | 16.1 | 8.2                           | 8.5 | 14.2 | 12.2 | 10.4 | 18.6 | 33.7                  | 27.1 | 30.4 | 27.2 | 12.5                  | 0    | 0.551 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.4                   | 91.7            | 70.1 | 89.2 | 69.5 | 86.1 | 69.1 | 75.1                | 83.0 | 73.3 | 81.8 | 71.8 | 80.4 | 11.0                 | 80   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.2                            | 125.1 | 78.1 | 71.3 | 116.9 | 76.6 | 69.5 | 109.8 | 75.3 | 38.8          | 83.8 | 37.2 | 81.4 | 35.8 | 79.4 | 82.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 25.1              | 21.4 | 19.0 | 5.1                        | 96.6 | 4.4                | 3.3 | 2.0                                                | 99.0 | -0.6       | 100.9 | -3.1       | 102.8 | -6.4       | 105.2 |
|                   |      |      | 4.6                        | 78.1 | 4.4                | 2.3 | 1.4                                                | 79.8 | -1.2       | 81.1  | -3.7       | 82.4  | -6.9       | 84.1  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 53.8   | 32.3  | 36.0  | 43.2  | 51.6  | 61.7  | 69.9  | 74.5  | 74.2  | 65.5  | 54.9  | 45.5  | 35.3 |
|                                                                                  | DBStd       | 16.37  | 9.03  | 9.24  | 7.57  | 6.42  | 5.93  | 5.18  | 4.44  | 5.09  | 5.98  | 7.44  | 8.47  | 8.42 |
|                                                                                  | HDD50       | 1915   | 548   | 396   | 235   | 53    | 2     | 0     | 0     | 0     | 0     | 38    | 185   | 458  |
|                                                                                  | HDD65       | 4948   | 1012  | 812   | 676   | 403   | 135   | 16    | 1     | 2     | 67    | 319   | 586   | 919  |
|                                                                                  | CDD50       | 3311   | 0     | 4     | 24    | 102   | 365   | 596   | 759   | 751   | 466   | 191   | 49    | 4    |
|                                                                                  | CDD65       | 871    | 0     | 0     | 0     | 2     | 33    | 162   | 295   | 289   | 82    | 8     | 0     | 0    |
|                                                                                  | CDH74       | 7874   | 0     | 0     | 3     | 37    | 296   | 1425  | 2761  | 2705  | 586   | 59    | 2     | 0    |
|                                                                                  | CDH80       | 2777   | 0     | 0     | 0     | 7     | 47    | 442   | 1061  | 1085  | 129   | 6     | 0     | 0    |
| Wind                                                                             | WSAvg       | 9.5    | 10.7  | 10.7  | 10.6  | 9.8   | 8.8   | 8.5   | 8.5   | 8.5   | 9.1   | 9.2   | 9.4   | 10.4 |
| Precipitation                                                                    | PrecAvg     | 18.8   | 1.2   | 0.8   | 1.3   | 1.4   | 2.0   | 2.2   | 2.0   | 1.5   | 2.0   | 1.5   | 1.3   | 1.5  |
|                                                                                  | PrecMax     | 25.0   | 2.9   | 2.4   | 3.0   | 3.9   | 5.0   | 5.1   | 5.3   | 5.0   | 4.5   | 3.0   | 2.8   | 3.6  |
|                                                                                  | PrecMin     | 10.1   | 0.1   | 0.1   | 0.0   | 0.1   | 0.5   | 0.6   | 0.3   | 0.2   | 0.2   | 0.1   | 0.0   | 0.1  |
|                                                                                  | PrecStd     | 4.0    | 0.8   | 0.6   | 0.7   | 0.8   | 1.2   | 1.1   | 1.3   | 1.3   | 1.4   | 0.8   | 0.8   | 0.9  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 56.9  | 64.5  | 71.4  | 78.8  | 84.5  | 91.4  | 96.9  | 96.7  | 89.2  | 80.4  | 71.4  | 59.4 |
|                                                                                  |             | MCWB   | 50.8  | 53.2  | 54.2  | 60.2  | 65.5  | 71.4  | 69.2  | 70.4  | 67.3  | 64.4  | 60.6  | 54.2 |
|                                                                                  | 2%          | DB     | 50.3  | 57.4  | 65.8  | 71.8  | 80.6  | 87.9  | 91.6  | 92.9  | 83.9  | 74.9  | 64.6  | 53.9 |
|                                                                                  |             | MCWB   | 47.0  | 50.9  | 52.6  | 58.5  | 65.3  | 69.1  | 70.0  | 69.4  | 65.9  | 62.6  | 58.4  | 50.4 |
|                                                                                  | 5%          | DB     | 46.5  | 52.0  | 60.5  | 67.6  | 77.1  | 85.6  | 88.2  | 89.5  | 80.4  | 70.0  | 60.5  | 50.2 |
|                                                                                  |             | MCWB   | 45.1  | 47.4  | 50.8  | 57.1  | 63.5  | 68.8  | 70.0  | 69.6  | 65.6  | 61.4  | 56.6  | 48.5 |
|                                                                                  | 10%         | DB     | 43.1  | 46.8  | 55.3  | 63.0  | 73.8  | 82.3  | 85.9  | 86.1  | 77.1  | 66.2  | 57.0  | 46.6 |
|                                                                                  |             | MCWB   | 42.2  | 44.5  | 48.1  | 54.2  | 62.1  | 67.5  | 69.7  | 69.1  | 65.2  | 59.2  | 54.2  | 45.4 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 51.7  | 56.0  | 57.1  | 64.3  | 70.6  | 75.2  | 77.2  | 78.9  | 73.3  | 68.7  | 62.6  | 55.1 |
|                                                                                  |             | MCDB   | 55.6  | 60.7  | 67.2  | 74.0  | 78.0  | 83.4  | 83.8  | 85.9  | 78.2  | 74.0  | 67.2  | 58.3 |
|                                                                                  | 2%          | WB     | 47.7  | 51.5  | 54.0  | 60.5  | 68.0  | 72.9  | 74.8  | 75.5  | 70.7  | 65.3  | 59.8  | 51.4 |
|                                                                                  |             | MCDB   | 49.9  | 56.7  | 63.0  | 67.7  | 76.0  | 81.4  | 83.0  | 83.4  | 76.8  | 70.4  | 64.0  | 53.9 |
|                                                                                  | 5%          | WB     | 44.8  | 48.1  | 51.3  | 58.0  | 65.7  | 71.0  | 73.2  | 73.4  | 68.7  | 62.9  | 57.2  | 48.5 |
|                                                                                  |             | MCDB   | 46.1  | 51.7  | 58.9  | 65.6  | 73.8  | 79.7  | 82.1  | 82.0  | 75.1  | 67.9  | 59.8  | 50.1 |
|                                                                                  | 10%         | WB     | 42.4  | 44.8  | 48.8  | 55.7  | 63.7  | 69.5  | 71.7  | 71.5  | 66.9  | 60.6  | 54.7  | 45.4 |
|                                                                                  |             | MCDB   | 43.3  | 46.7  | 54.2  | 61.7  | 71.0  | 78.6  | 80.9  | 80.3  | 73.7  | 65.4  | 56.9  | 46.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.4  | 13.9  | 17.1  | 18.4  | 19.3  | 20.0  | 20.4  | 20.6  | 18.9  | 16.6  | 13.4  | 12.0 |
|                                                                                  |             | MCDBR  | 16.7  | 21.9  | 25.8  | 25.0  | 24.0  | 23.9  | 23.6  | 24.3  | 23.0  | 21.9  | 19.1  | 16.8 |
|                                                                                  |             | MCWBR  | 13.2  | 15.3  | 15.0  | 14.2  | 11.9  | 10.8  | 10.0  | 10.4  | 10.6  | 12.2  | 12.9  | 13.5 |
|                                                                                  | 5% WB       | MCDBR  | 15.7  | 20.5  | 23.2  | 21.8  | 21.2  | 20.9  | 20.2  | 20.8  | 19.3  | 19.4  | 17.0  | 16.1 |
|                                                                                  |             | MCWBR  | 13.0  | 15.3  | 14.8  | 14.0  | 12.3  | 10.9  | 10.3  | 10.9  | 11.1  | 12.7  | 12.5  | 13.5 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.345  | 0.373 | 0.404 | 0.439 | 0.416 | 0.418 | 0.431 | 0.441 | 0.416 | 0.387 | 0.364 | 0.345 |      |
|                                                                                  | taud        | 2.320  | 2.252 | 2.202 | 2.126 | 2.275 | 2.301 | 2.275 | 2.239 | 2.281 | 2.333 | 2.339 | 2.338 |      |
|                                                                                  | Ebn at Noon | 239    | 253   | 261   | 261   | 272   | 271   | 266   | 258   | 254   | 246   | 230   | 226   |      |
|                                                                                  | Edh at Noon | 26     | 33    | 40    | 47    | 42    | 41    | 41    | 41    | 36    | 30    | 26    | 23    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 414    | 668   | 1054  | 1517  | 1917  | 2085  | 2083  | 1864  | 1350  | 848   | 478   | 364   |      |
|                                                                                  | RadStd      | 59     | 89    | 107   | 150   | 117   | 118   | 108   | 101   | 103   | 80    | 60    | 52    |      |

## Historical Trends

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            |  | +0.88 | N/A     | N/A    | N/A     | N/A   | -1.39 | N/A         | N/A   | +207  | +133  |
| Regional (31 neighbors) |  | +0.88 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -220  | +160  | +103  |

Nomenclature: See separate page